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Buatlah 1 contoh tata bahasa reguler (regular grammar), dari contoh tersebut tentukan :

- Bahasa reguler (regular language)
- Eksresi reguler (regular expression)
- Mesin finite State Automata

Jawab!

Regular grammar :

$$G = (V, T, S, P)$$

$$V = \{S, A, B, V_f\}$$

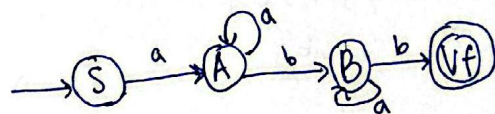
$$T = \{A, b\}$$

$$S = S$$

$$P = S \rightarrow aA$$

$$A \rightarrow aA \mid bB$$

$$B \rightarrow aB \mid bV_f$$



a) Regular language (Bahasa reguler)

String Yang diterima : $\{abb, aabb, aaabb, aabab, aaabab, \dots\}$

String Yang ditolak : $\{a, Ab, bba, aba, \dots\}$

$$L(G) = \{abb\} \cup \{aabb\} \cup \{aaabb\} \cup \{ab \dots b\}$$

$$L(G) = \{a^n b a^m b \mid n \geq 1, m \geq 0\}$$

↳ diawali dengan satu atau lebih a, lalu satu b, boleh ada beberapa a lagi, diakhiri b

b) Ekspresi reguler (regular expression)

$$ER = a^+ b a^* b$$

$$ER = (a^+ b b) \cup (a^+ b a^+ b)$$

c) Mesin finite State Automata

- Non-deterministic finite Automata (NFA)

$$M = (Q, \Sigma, \delta, S, F) \quad S = q_0, A = q_1, B = q_2, V_f = q_3$$

$$Q = \{q_0, q_1, q_2, q_3\}$$

$$\Sigma = \{a, b\}$$

$$F = \{q_3\}$$

$$S = q_0$$

$$\delta = \delta(q_0, a) = q_1$$

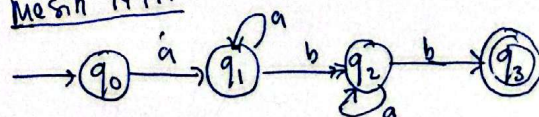
$$\delta(q_1, a) = q_1$$

$$\delta(q_1, b) = q_2$$

$$\delta(q_2, a) = q_2$$

$$\delta(q_2, b) = q_3$$

Mesin NFA :



δ	a	b
q ₀	q ₁	-
q ₁	q ₁	q ₂
q ₂	q ₂	q ₃
q ₃	-	-